

Day - 18

PERCENTAGE Based on Salary

1) Radha spends 40% of her salary on food, 20% on house rent, 10% on entertainment and 10% on conveyance. If her saving at the end of month are Rs 1500, then her salary per month (in Rs.) is

$$100\% - 40\% - 20\% - 10\% - 10\%$$

$$80\% = 80\%$$

$$20\% = 1500$$

$$100\% = x$$

$$20x = 1500 \times 100$$

$$x = 7500$$

2) kishan spends 30% of his salary on food and donates 3% in a charitable trust. He spends RS. 2310 on these two items, then total salary for that month is

$$30\% + 3\%$$

$$33\% = 2310$$

$$100\% = x$$

$$33x = \overset{70}{2310} \times 100 \Rightarrow 7000$$

3) Mr. x spends 35% of his salary on food 5% of his salary on children education.

In January 2011, he spend RS. 17600 on these two items. His salary for that month is

$$35\% + 5\%$$

$$x = 44,000$$

$$40\% = 17600$$

$$100\% = x$$

$$x = \frac{17600 \times 100}{40} = 44,000$$

4) Keshav spent RS. 55475 on his birthday party RS. 28525 on buying home appliances and the remaining 25% of the total amount he had as cash with him. What was the total amount?

$$55475 + 28525 = 84,000$$

$$75\% = 84,000$$

$$100\% = x$$

$$75x = \frac{84,000}{112} \times 100$$

$$x = 112000$$

5) Ms. Sujata invests 7% i.e., RS. 2100 of her monthly salary in mutual fund. Later she invests 18% of her monthly salary in recurring deposits. Also, she invests 6%

of her salary on nsc's. What is
the total annual amount invested
by ms. Su?

$$\begin{array}{l} 7\% = 2170 \\ 100\% = x \end{array} \quad \left. \vphantom{\begin{array}{l} 7\% = 2170 \\ 100\% = x \end{array}} \right\} 7x = 2170 \times 100 = 217000$$

$$\boxed{100\%} - 7\% + 18\% + 6\%$$

$$= 31\%$$

$$100\% = 31,000$$

$$31\% = x$$

$$x = 31,000 \times 31$$

$$x = 9610$$

$$x = 9610$$

$$x \times 12 = 9610 \times 12$$

$$= 115320$$

6) Mr. X spends 20% of household out of the remaining 25% he spends on children's education, 15% on transport, 15% on medicine and 10% on entertainment. He is left with Rs. 9800 after incurring all these expenditures. What is his monthly income?

Rs 100.

$$\rightarrow 20\% \text{ of } 100 = 20 \text{ (HT)}$$

$$R: (60)$$

$$25\% + 15\% + 15\% + 10\%$$

$$65\%$$

13

$$486 \times \frac{65}{100} \Rightarrow 52 \Rightarrow 82 - 52$$

%

$$\Rightarrow 28\%$$

$$28\% = 9800$$

$$100\% = x$$

$$\Rightarrow 28x = 9800 \times 100$$

$$x \Rightarrow 3500.$$

7) A man spends 40% of his monthly salary on food and one-third of the remaining on transport. If he saves Rs. 4500 per month, which is equal to half the balance after spending on food and transport, his monthly salary is

$$\text{Food } 40\% \text{ of } 100 = (40)$$

Rem

$$60$$

$$\text{Trans } 60 \times \frac{1}{3} \Rightarrow 40 = \text{rem}$$

$$20\% = 4500$$

$$x = \frac{4500 \times 100}{20}$$

$$100\% = x$$

$$x = 22500$$

8) A person gave 20% of his income to his elder son, 30% of the remaining to the younger son and 10% of the balance, he donated to a trust. He is left with Rs. 10000. His income was:

$$\underline{100}$$

$$\hookrightarrow 20\% \text{ of } 100 = 20\%$$

$$\underline{80}$$

$$30\% \text{ of } 80 = 24$$

$$\underline{56}$$

$$\hookrightarrow 10\% \text{ of } 56 = 5.6$$

$$50.4\% = 10,080$$

$$100\% = x$$

$$x = 20000$$

9) The monthly salaries of A and B together amount to RS 40000. A spends 85% of his salary, and B, 95% of his salary. If now their savings are the same, then the salary of A is

A	B
x	$(40,000 - x)$

$$S_1 = S_2$$

~~x~~

$$x \times \frac{15}{100} = (40,000 - x) \times \frac{5}{100}$$

$$x + 3x = 40,000$$

$$4x = 40,000$$

$$x = 10,000$$

$$y = 30,000$$